



Tri-Organism Geometry: A Formal Interaction Framework for Lume-Native Physical Flow Organisms

Deterministic 2D/3D Geometry, Pairwise Interaction Channels, and Cross-Node Coupling for the Meridian / Verdara Ultra / HydroCore Organism System

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Companion Papers (required reading): Meridian Architecture · Meridian as Synthetic Organism · Verdara Ultra · HydroCore

Related: Lume [10.5281/zenodo.19382282](https://zenodo.org/record/19382282) · Trust Layer [10.5281/zenodo.19560674](https://zenodo.org/record/19560674) · DAIGS [10.5281/zenodo.19491784](https://zenodo.org/record/19491784)

Lume-V [10.5281/zenodo.19645097](https://zenodo.org/record/19645097) · Lume-X [10.5281/zenodo.19443968](https://zenodo.org/record/19443968) · Dissolution [10.5281/zenodo.15065493](https://zenodo.org/record/15065493)

Patent Pending — DarkWave Studios LLC
Systems and Methods for Deterministic Multi-Layer Wireless Energy Routing

Lume-X: Provisionally patented deterministic control runtime
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Abstract

I present the Tri-Organism Geometry — a formal framework for the two-dimensional and three-dimensional visual representation of, and deterministic interaction modeling between, three Lume-native physical flow organisms: Meridian (energy domain), Verdara Ultra (outdoor domain), and HydroCore (hydrological domain). Each organism implements the Lume 4/42 architecture independently, maintaining its own four primitive flows and forty-two operational nodes. The Tri-Organism Geometry defines how these three organisms are co-located in a shared geometric frame, how their pairwise interactions are encoded visually and computationally, and how their combined state is synthesized into a central meta-core that provides a single composite view of tri-organism equilibrium.

The framework makes four primary architectural contributions. First, it defines a triangular organism layout — an equilateral arrangement with Meridian at the apex, Verdara Ultra at the lower left, and HydroCore at the lower right — that preserves each organism's full 4/42 internal structure while situating them in a shared coordinate system. Second, it specifies three pairwise interaction channels — Meridian↔Verdara Ultra (energy-outdoor), Verdara Ultra↔HydroCore (outdoor-water), and HydroCore↔Meridian (water-energy) — as directed, bidirectional conduits encoding the physical coupling relationships between domains. Third, it defines a cross-node linking system that identifies specific nodes across different organisms whose states are physically coupled and renders those couplings as optional visual overlays. Fourth, it specifies a central meta-core — a small three-axis composite organism at the centroid of the triangle — that encodes the aggregate health of all three organisms in a single glanceable structure.

A three-dimensional extension stacks the three organisms as layered planes — Meridian above (infrastructure layer), Verdara Ultra at center (surface layer), HydroCore below (subsurface layer) — with tetrahedral interaction geometry connecting them across all three spatial dimensions. Alert propagation across organism boundaries — where a critical node in one organism triggers visual signaling on the connecting edge and distortion of the central meta-core — is formally specified, enabling operators and practitioners to immediately identify cascades that originate in one domain and propagate into another.

The Tri-Organism Geometry is the first published framework for multi-organism interaction geometry in the Lume ecosystem. It establishes the foundational visual and computational language for systems in which multiple Lume organisms must coexist, interact, and be simultaneously governed in real time.

Keywords: multi-organism geometry, Lume synthetic organism, deterministic interaction framework, physical flow organisms, tri-organism system, cross-domain coupling, Meridian, Verdara Ultra, HydroCore, organism visualization, meta-core, interaction channels

1 Introduction

1.1 The Multi-Organism Problem

The Lume ecosystem has produced a family of physical flow organisms, each governing a distinct domain of continuous physical reality. Meridian governs energy flow. Verdara Ultra governs outdoor environmental flow. HydroCore governs hydrological flow. BioCore governs internal biological flow. Each organism is formally specified, internally coherent, and capable of deterministic autonomous operation within its domain.

But the physical world does not honor domain boundaries. Energy systems depend on water. Water systems depend on terrain and weather. Outdoor safety depends on energy availability and water access. In any real physical deployment — a mountain community, a wilderness expedition with communications infrastructure, a regional watershed that includes hydroelectric generation — these organisms are not merely simultaneous. They are coupled. The state of one organism influences the state of another in ways that are physically real, causally grounded, and operationally significant.

A framework that visualizes and governs three or more Lume organisms in isolation — each in its own interface, each producing its own recommendations without knowledge of the others — misses the most important information: the cross-domain interactions that can turn a local stress in one organism into a cascade failure across multiple domains. The energy system that loses generation capacity due to drought (HydroCore → Meridian). The outdoor routing system that must account for flooded river crossings caused by upstream hydrological events (HydroCore → Verdara Ultra). The wilderness water source that is contaminated by a weather event that also closes the planned evacuation route (both HydroCore and Verdara Ultra simultaneously).

The Tri-Organism Geometry is the first Lume framework specification to address this multi-organism problem formally.

1.2 The Geometry Reframe

The central insight of this work is that multi-organism interaction is fundamentally a geometric problem before it is a computational one. Each Lume organism is a geometric object — a 4-axis core surrounded by a 42-node ring in a deterministic polar coordinate system. When multiple organisms are placed in a shared geometric frame, the spatial relationships between them become the visual encoding of their real-world physical relationships.

The geometry is not decorative. Placing Meridian at the apex of a triangle, Verdara Ultra at the lower left, and HydroCore at the lower right is not an arbitrary aesthetic choice. It encodes the physical layering of the domains: energy infrastructure sits above the surface world (aerial towers, transmission lines, solar installations); outdoor terrain and weather occupy the surface where human movement occurs; water flows beneath and through the surface in hydrological networks. The triangular arrangement makes this stratification immediately legible.

The edges of the triangle are interaction channels — physically grounded conduits through which state information flows between organisms. The width, color, and directionality of flow within each channel encodes the actual physical coupling strength between the domains it connects. The central meta-core at the centroid of the triangle is the system-level awareness point — the one structure that always tells you whether the tri-organism system as a whole is in equilibrium or in distress.

1.3 Novel Contributions

This work makes six distinct contributions not present in existing multi-system visualization or multi-agent architecture literature:

1. **The first formal multi-organism geometry specification** in the Lume ecosystem — establishing coordinate system, layout conventions, and rendering rules for co-locating multiple 4/42 organisms in a shared geometric frame.
2. **Pairwise interaction channel formalism** — defining interaction channels as first-class geometric and computational objects with direction, magnitude, semantic grounding, and visual encoding rules.
3. **Cross-node coupling taxonomy** — a complete formal table of the specific node-to-node couplings between organisms, grounded in the physical relationships

between energy, outdoor, and hydrological domains.

4. **Central meta-core specification** — a three-axis composite organism at the system centroid that synthesizes tri-organism health into a single glanceable structure while remaining directly driven by live organism state vectors.
5. **Three-dimensional tetrahedral extension** — a 3D geometry that adds vertical domain stratification and tetrahedral interaction geometry connecting organisms across all three spatial dimensions.
6. **Alert cascade propagation model** — a formal specification of how critical node states propagate visually and computationally across organism boundaries, enabling immediate identification of cross-domain cascades.

1.4 Scope

This paper specifies the geometry and interaction framework for three organisms: Meridian, Verdara Ultra, and HydroCore. BioCore — the internal biological flow organism — is not included in the Tri-Organism Geometry because its domain (internal human biology) is not physically co-located with the external infrastructure, terrain, and water domains of the other three. BioCore interacts with the other organisms differently — through the human practitioner who is embedded in the physical world those organisms govern — and its multi-organism integration framework is a subject for a separate specification.

The geometry defined here is extensible to additional physical domain organisms as the Lume ecosystem grows. The triangular three-organism layout is the initial instantiation of a more general N-organism polygon geometry.

1.5 Paper Organization

Section 2 establishes background on the three organisms and their physical relationships. Section 3 defines the triangular layout. Section 4 specifies individual organism geometry within the shared frame. Section 5 defines the pairwise interaction channels. Section 6 defines cross-node coupling links. Section 7 specifies the central meta-core. Section 8 presents the 3D extension. Section 9 defines alert propagation and cascade visualization. Section 10 specifies the user interaction model. Section 11 addresses implementation architecture. Section 12 discusses applications and future extensions.

2 Background — The Three Organisms and Their Physical Relationships

2.1 Meridian — Energy Domain

Meridian is a four-layer deterministic wireless energy routing architecture implemented as a Lume synthetic organism [Andrews, 2026]. In the Tri-Organism Geometry, Meridian is represented by its 4/42 organism structure: four energy flow primitives and forty-two energy state nodes. Meridian governs the availability, routing, and safe delivery of electrical energy across infrastructure networks.

In the real-world contexts most relevant to the Tri-Organism Geometry — remote infrastructure, wilderness-adjacent communities, outdoor expedition operations — Meridian's energy state directly constrains the operational capabilities of the other two organisms. Communication systems, navigation devices, weather monitoring stations, water pumping systems, and lighting all depend on energy availability. When Meridian's state degrades, the operational envelope of Verdara Ultra and HydroCore narrows — not because of any failure in their own domains, but because the infrastructure layer they depend on has become constrained.

2.2 Verdara Ultra — Outdoor Domain

Verdara Ultra is a four-primitive deterministic outdoor flow organism for terrain, weather, environmental, and biological flow governance [Andrews, 2026]. In the Tri-Organism Geometry, Verdara Ultra is represented by its 4/42 organism structure: four outdoor flow primitives and forty-two environmental state nodes.

Verdara Ultra governs the outdoor physical environment through which humans and infrastructure must operate. Its state determines routing safety, weather exposure, terrain traversability, and biological hazard. In the Tri-Organism Geometry context, Verdara Ultra is the surface organism — it occupies the domain that both Meridian (infrastructure built on and above the surface) and HydroCore (water that flows across and beneath the surface) must interact with.

2.3 HydroCore — Hydrological Domain

HydroCore is a four-primitive deterministic hydrological flow organism for pressure, volume, purity, and thermal flow governance [Andrews, 2026]. In the Tri-Organism Geometry, HydroCore is represented by its 4/42 organism structure: four hydrological

flow primitives and forty-two water state nodes.

HydroCore governs water systems from individual pipes to regional watersheds. Its state determines water availability, purity, flood and drought risk, and infrastructure water-dependency. In the Tri-Organism Geometry context, HydroCore is the subsurface organism — it governs what moves beneath the terrain that Verdara Ultra monitors and through the infrastructure that Meridian powers.

2.4 Physical Coupling — Why These Three Organisms Interact

The physical coupling relationships between these three organisms are not abstract. They are the governing physics of the real systems they represent.

Energy ↔ Water (Meridian ↔ HydroCore): Hydroelectric generation — micro-hydro and grid-scale hydro — directly couples HydroCore’s flow rate (V1), head height (P4), and hydraulic gradient (P5) to Meridian’s generation capacity. When water flow drops in a drought (HydroCore V8 at caution), hydro generation drops, and Meridian’s available energy capacity contracts. In the reverse direction, water pumping, treatment, and distribution systems are major energy consumers — a Meridian outage reduces pump capacity, which reduces HydroCore’s ability to maintain pressure and distribution continuity.

Outdoor ↔ Water (Verdara Ultra ↔ HydroCore): Terrain saturation, storm events, and snowmelt drive HydroCore’s flood probability (V7). A Verdara Ultra storm event (W4, W5 at caution) is simultaneously a HydroCore flood precursor. Conversely, drought conditions (HydroCore V8 at critical) drive trail closures, wildlife behavior changes, and reduced water source availability for outdoor movement — all feeding back into Verdara Ultra’s biological and terrain nodes.

Energy ↔ Outdoor (Meridian ↔ Verdara Ultra): Outdoor infrastructure — weather monitoring stations, communication repeaters, navigation beacon networks, search and rescue communication systems — depends on Meridian’s energy availability. When Meridian’s state degrades, Verdara Ultra loses access to real-time weather data and communication support, which degrades its ability to maintain accurate W9 (Weather Volatility) and W10 (Safe Weather Window) estimates. Conversely, extreme weather events (W2, W4, W5 at critical in Verdara Ultra) impose direct physical stress on outdoor energy infrastructure — downed transmission lines, iced solar panels, storm-damaged relay stations — feeding back into Meridian’s node states.

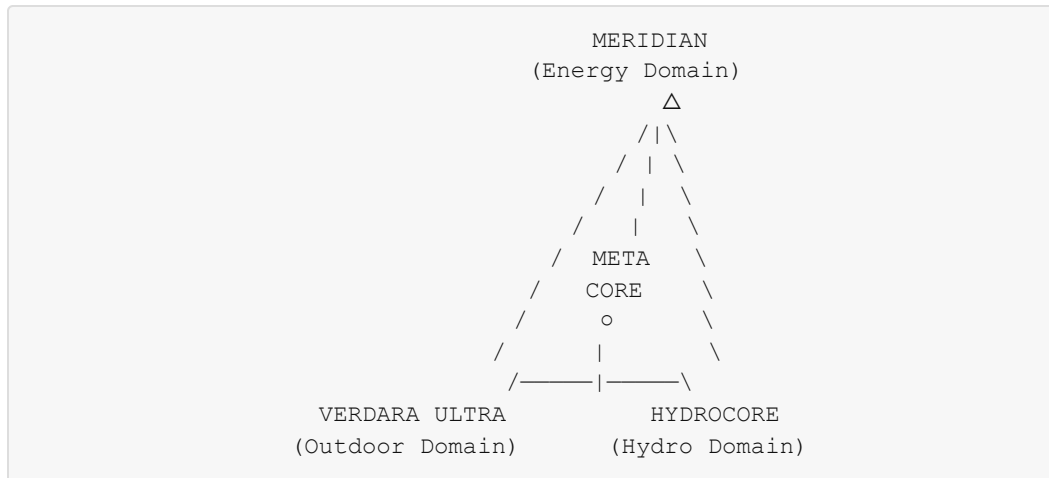
3 The Triangular Organism Layout

3.1 Layout Philosophy

The triangular layout is the minimal geometric arrangement that: maintains each organism as a distinct visual entity with full internal structure; establishes pairwise relationships between every pair of organisms; creates a central region that can host the meta-core without overlapping any organism; and supports natural scale and semantic interpretation (vertical height = domain layer). An equilateral triangle guarantees that all three organisms are equidistant from each other and from the central meta-core, encoding the principle that no organism has a privileged structural position relative to the others.

3.2 Vertex Assignments

The three organisms are assigned to the three vertices of the equilateral triangle according to their domain stratification:



- **Meridian — Apex (top):** Energy infrastructure occupies the highest visual position — encoding its role as the infrastructure layer that sits above and powers the surface and subsurface domains.
- **Verdara Ultra — Lower left:** The outdoor surface domain. In the 3D extension, this is the middle layer.
- **HydroCore — Lower right:** The hydrological subsurface domain. In the 3D extension, water systems exist beneath and through the terrain that Verdara Ultra governs.

3.3 Coordinate System

The shared coordinate system for the Tri-Organism Geometry is defined as follows:

```
Origin: centroid of the equilateral triangle
Triangle side length: L (implementation parameter)
Vertex positions:
  Meridian:      (0, L/√3)           ≈ (0, 0.577L)
  Verdara Ultra: (-L/2, -L/(2√3))   ≈ (-0.5L, -0.289L)
  HydroCore:     (+L/2, -L/(2√3))   ≈ (+0.5L, -0.289L)
```

Each organism's 4/42 core is centered on its vertex. Each organism's ring has a radius appropriate to fill roughly one-third of its vertex region without overlapping the triangle edges.

3.4 Minimum Separation Constraint

The rings of adjacent organisms must not overlap. The minimum allowable ring radius R_{ring} for each organism is:

$$R_{\text{ring}} \leq (L/2 - \text{edge_clearance}) \times \sin(60^\circ)$$

In practice, the organism ring radii are set to approximately 30% of L, leaving 20% of L as the interaction channel width along each edge.

4 Individual Organism Geometry Within the Shared Frame

4.1 Preserved Internal Structure

Each organism retains its complete internal 4/42 geometry within the shared frame. No organism's internal structure is modified, simplified, or aggregated for the purpose of the Tri-Organism Geometry. The geometry adds a shared frame around fully-specified organisms; it does not reduce them.

This preservation principle is architecturally important: an operator who needs to examine Meridian's internal node states can do so within the Tri-Organism frame with exactly the same level of detail as in Meridian's standalone geometry. The Tri-Organism view is additive — it adds inter-organism information without removing intra-organism information.

4.2 Color Identity Preservation

Each organism's color palette is preserved from its standalone specification:

- **Meridian:** Electric blues, violets, whites — the colors of electromagnetic energy
- **Verdara Ultra:** Earth browns, forest greens, sky blues — the colors of outdoor terrain, vegetation, and sky
- **HydroCore:** Deep blues, cyans, whites — the colors of water at depth and in motion

Color identity serves as the primary visual signal for origin attribution throughout the Tri-Organism Geometry. When flow lines, interaction channels, cross-node links, and meta-core axes are colored, the color always indicates which organism is the source or which domain the information belongs to. Color is never decorative in this geometry.

4.3 Scale Consistency

All three organisms are rendered at the same scale within the Tri-Organism frame, with equal ring radii and equal axis arm lengths at their default (neutral) states. This equal-scale rendering encodes the principle that no organism is more important than another in the shared frame — importance is encoded by state (a critically degraded organism is visually prominent regardless of which organism it is), not by size.

4.4 Internal Detail Level

The Tri-Organism frame supports three levels of internal organism detail:

- **Overview level:** Only the four primitive axes and the ring envelope (the contour connecting all node positions) are rendered. Individual nodes are not visible. Maximizes readability at the tri-organism scale.
- **Standard level:** The four primitive axes, the ring envelope, and individual node dots with state-colored fill are rendered. Intra-primitive arc connections are shown. This is the default operating level.
- **Detail level:** Full node labeling, threshold indicators, and numeric state values are shown for the selected organism. The other two organisms revert to overview level to reduce visual noise.

5 Pairwise Interaction Channels

5.1 Channel Definition

A pairwise interaction channel is a directed, bidirectional geometric conduit connecting two organisms along one edge of the equilateral triangle. Each channel encodes the physical coupling relationship between its two organisms through: geometry (a semi-transparent band spanning the triangle edge), direction (directional flow lines indicating current influence), magnitude (line thickness encoding strength), and color (source organism color encoding origin).

Channel	Edge	Physical Relationship
Meridian ↔ Verdara Ultra	Apex–Left	Energy infrastructure ↔ Outdoor environment
Verdara Ultra ↔ HydroCore	Left–Right	Outdoor terrain/weather ↔ Water systems
HydroCore ↔ Meridian	Right–Apex	Water systems ↔ Energy infrastructure

5.2 Meridian ↔ Verdara Ultra Channel

Physical grounding: This channel represents the energy-outdoor coupling — the dependence of outdoor operational infrastructure on energy availability, and the effect of outdoor conditions on energy infrastructure.

Meridian → Verdara Ultra (energy supplies outdoor): Weather station network availability (powered monitoring), communication repeater network coverage, navigation beacon network availability, and emergency lighting/shelter systems. When Meridian’s state is degraded: Verdara Ultra’s W9 (Weather Volatility) and W10 (Safe Weather Window) estimates become less accurate; B11 (Rescue Accessibility) is elevated; the routing engine’s information quality decreases, triggering automatic conservatism multipliers.

Verdara Ultra → Meridian (outdoor conditions stress energy): Extreme weather events (W2 Wind, W4 Precipitation, W6 Storm) impose physical stress on exposed energy infrastructure. High wildfire risk threatens ground-based energy infrastructure. Terrain accessibility constraints (T5, T7) slow maintenance response to Meridian failures.

5.3 Verdara Ultra ↔ HydroCore Channel

Physical grounding: This channel represents the outdoor-water coupling — how terrain and weather conditions drive hydrological events, and how water availability and hazard affect outdoor safety.

Verdara Ultra → HydroCore (outdoor conditions drive hydrology): Storm intensity (W4, W5) drives surface runoff, flash flood risk (V7), and water quality degradation (U1, U2). Snow cover and melt rate drive seasonal flow rate (V1). Terrain saturation (T4 + W5) drives Soil Absorption (V5) and Flood Probability (V7). Wildfire burn areas dramatically reduce V5 and amplify contamination risk.

HydroCore → Verdara Ultra (hydrology shapes outdoor conditions): Drought (V8) drives elevated fire risk, reduced vegetation (B5), and reduced water availability (B7). Flood events (V7 at critical) close terrain crossings (T9). Water quality events (U4 Pathogen) elevate B4 (Pathogen Risk). Ice formation (T3 Freeze) drives T8 (Traction Quality) degradation.

5.4 HydroCore ↔ Meridian Channel

Physical grounding: This channel represents the water-energy coupling — the most direct infrastructure dependency in the trio, present in virtually every real-world deployment context where hydro generation or water-dependent energy infrastructure exists.

HydroCore → Meridian (water drives energy): Hydroelectric generation capacity is directly coupled to V1 (Flow Rate), P4 (Head Height), and P5 (Hydraulic Gradient). Drought (V8) reduces hydro generation. Water availability for cooling constrains thermal plant capacity. Floods (V7) can damage energy infrastructure directly.

Meridian → HydroCore (energy constrains water systems): Water pumping/distribution systems are major energy consumers. Water treatment requires energy — Meridian degradation risks purity governance failure (U11). SCADA monitoring requires energy — Meridian failure causes HydroCore to lose sensor coverage.

6 Cross-Node Coupling Links

6.1 Definition

Cross-node coupling links are thin, semi-transparent lines connecting specific nodes in different organisms whose states are physically coupled — where a change in one node’s state produces a correlated change in the connected node in another organism. These links make the inter-organism coupling visible at the node level, not just at the primitive or organism level.

Cross-node links are rendered as optional overlays — they are not shown by default, but can be activated through the interaction model (Section 10). When active, only links with non-trivial coupling strength (both participating nodes at advisory or below) are shown.

6.2 Coupling Strength Encoding

Each cross-node link is rendered with: color (source organism color), opacity (coupling strength, 0.0–1.0), and animation (pulsing in the direction of influence). Coupling strength is computed as:

$$\text{strength} = \min(|s_{\text{source}}|, |s_{\text{target}}|) \times \text{correlation_coefficient}$$

Where *correlation_coefficient* is a domain-grounded constant in [0, 1] for each node pair (see Appendix B).

6.3 Primary Cross-Node Coupling Pairs

Meridian ↔ Verdara Ultra couplings:

Meridian Node	Verdara Ultra Node	Physical Relationship	Corr.
Generation capacity	W9 Weather Volatility	Weather data quality degrades with power loss	0.65
Load/outage risk	W10 Safe Weather Window	Comms loss reduces weather window accuracy	0.70
Distribution stability	B11 Rescue Accessibility	Power loss degrades rescue communication	0.80
Mesh fabric health	E4 Visibility (data)	Sensor network loss = degraded visibility data	0.55

Verdara Ultra ↔ HydroCore couplings:

Verdara Ultra Node	HydroCore Node	Physical Relationship	Corr.
W4 Precipitation Type	V7 Flood Probability	Storm precipitation drives flood risk	0.85
W5 Precipitation Intensity	V1 Flow Rate	Rainfall intensity drives flow increase	0.80
W5 Precipitation Intensity	U1 Turbidity	Rain events increase sediment/turbidity	0.75
T4 Surface Stability	V5 Soil Absorption	Saturated soils reduce absorption	0.70
W1 Temperature	T3 Freeze Risk	Air temperature drives water freeze risk	0.90
W1 Temperature	V8 Drought Probability	High temp accelerates evaporation/drought	0.65
B7 Water Availability	V8 Drought Probability	Drought reduces outdoor water access	0.75
B4 Pathogen Risk	U4 Pathogen Index	Surface water contamination events	0.70
T9 Water Crossing	V7 Flood Probability	Flood events make crossings hazardous	0.85

HydroCore ↔ Meridian couplings:

HydroCore Node	Meridian Node	Physical Relationship	Corr.
V1 Flow Rate	Generation capacity	Hydro generation is flow-dependent	0.90
P4 Head Height	Generation capacity	Head determines hydro generation potential	0.85
P5 Hydraulic Gradient	Generation capacity	Gradient determines flow velocity and power	0.80
V8 Drought Probability	Generation capacity	Drought directly threatens hydro generation	0.85
V7 Flood Probability	Mesh fabric health	Floods damage energy infrastructure	0.70
T3 Freeze Risk	Distribution stability	Ice formation on power lines/substations	0.60
U11 Potability Index	Mesh fabric health	Treatment system power requirement	0.65

7 The Central Meta-Core

7.1 Structure and Purpose

The central meta-core is a small composite organism located at the centroid of the equilateral triangle — the geometric center of the Tri-Organism frame. It serves one purpose: provide an immediate, glanceable assessment of the tri-organism system's overall health without requiring inspection of any individual organism.

The meta-core is not a separate organism. It does not have its own primitives, nodes, or governance logic. It is a visualization construct — a derived representation computed entirely from the current state vectors of the three organisms.

7.2 Meta-Core Structure

The meta-core consists of **three mini-axes** (one per organism, colored in that organism's primary color) and an **inner triangle** connecting the tips of the three axes:

- **Axis length** encodes the organism's global health scalar H , computed as the mean of its four primitive state scalars
- At full health ($H = +1.0$ for all three organisms), the three axes form an equilateral star

- As any organism's health degrades, its axis shortens; at critical health, the axis collapses toward the core center

```
Meridian axis:      color = electric violet, pointing up
Verdara Ultra axis: color = earth brown/green, pointing lower-left
HydroCore axis:     color = deep blue, pointing lower-right
```

The three axis directions match the directions from the meta-core centroid toward each organism's vertex — the geometry of the mini-axes mirrors the geometry of the full tri-organism triangle.

The **inner triangle** is formed by connecting the tips of the three mini-axes. Its fill color is a blended mix of all three organism colors, weighted by their current health scalars. When all three organisms are healthy, the triangle is a balanced, harmonious blend. When one organism degrades, its color contribution diminishes and the triangle distorts. When one organism reaches critical, the triangle loses a vertex — the most immediate visual signal in the entire Tri-Organism frame.

7.3 Meta-Core Health Computation

```
For each organism O:
     $H_O = (1/4) \times \sum(\text{primitive\_state\_i})$  for  $i = 1$  to 4

Meta-core axis length:
     $\text{axis\_length\_O} = \text{base\_length} \times (H_O + 1.0) / 2.0$ 

This maps  $H_O \in [-1.0, +1.0]$  to  $\text{axis\_length\_O} \in [0, \text{base\_length}]$ .
```

7.4 Meta-Core System State Classification

State	Condition	Visual Response
Tri-Equilibrium	All three H values $\geq +0.3$	Full-sized balanced inner triangle. No global alert. Standard monitoring.
Tri-Advisory	At least one H value in $[0.0, +0.3)$	Slight deformation. Advisory color cast toward degrading organism.
Tri-Caution	At least one H value in $[-0.5, 0.0)$	One axis noticeably shorter. Triangle visibly deformed. Cross-node coupling links auto-activate.
Tri-Critical	At least one $H \leq -0.5$, OR any node at critical	Axis collapsed or near-collapsed. Triangle severely deformed. Outer ring pulses. All channels highlighted.

8 Three-Dimensional Extension

8.1 3D Layout Philosophy

The two-dimensional layout is the primary operating view — it is the most information-dense rendering that is legible in real-time operation. The three-dimensional extension adds two capabilities that the 2D layout cannot provide: vertical domain stratification and temporal evolution visualization.

8.2 Vertical Domain Stratification

In the 3D view, the three organisms are separated along the Z-axis (vertical) according to their physical domain layer:

```
Z-axis assignments:
  Meridian:      +Z (high)  - energy infrastructure layer (aerial,
elevated)
  Verdara Ultra:  0 (center) - surface layer (terrain, weather,
outdoor)
  HydroCore:     -Z (low)   - subsurface layer (groundwater, piped
systems, riverbeds)
```

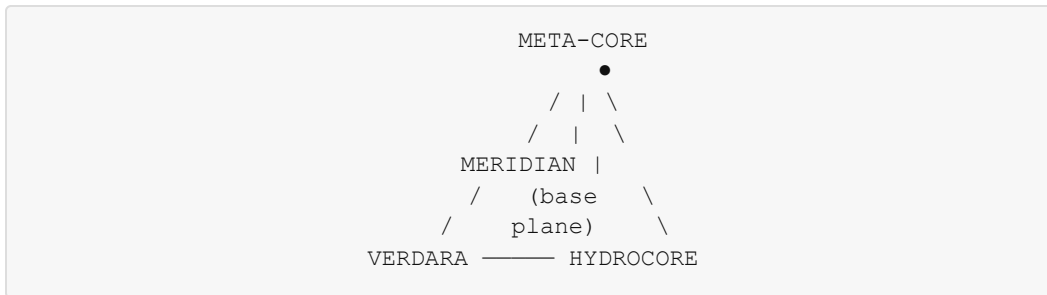
Each organism’s 4/42 geometry exists in its own horizontal plane at its assigned Z-height. The three planes are separated by a gap that is a rendering parameter.

8.3 Vertical Pillars

In the 3D stratified view, the pairwise interaction channels become vertical and diagonal conduits connecting the organism planes, forming a helical or triangular-prism structure in 3D.

8.4 Tetrahedral Option

An alternative 3D layout places the meta-core at a fourth point — elevated above the centroid of the three organism planes — forming a tetrahedron:



In this layout, the three edges of the tetrahedral base represent the pairwise organism interactions. The three edges from each organism to the apex represent each organism's contribution to the meta-core. This layout is most useful for executive-level system health visualization.

8.5 Temporal Extension

In temporal mode, each organism plane is extruded along the time axis (a fourth dimension rendered by stacking Z-slices with increasing time offset). This creates three parallel “organism history columns” — one per organism — each showing the evolution of that organism's 4/42 state over time. Temporal cross-domain events — where a state change in one organism is followed by a correlated change in another — are highlighted by connecting lines between corresponding Z-slices.

9 Alert Propagation and Cascade Visualization

9.1 Alert Types

Type	Condition	Rendering
Intra-organism	A node or primitive in a single organism crosses a threshold	That organism’s standard alert visualization only. No tri-organism response.
Cross-organism	Critical state in one organism triggers correlated degradation in a coupled node in another	Originating organism alert + connecting channel edge pulses + specific cross-node link highlighted.
System	Multiple organisms critical simultaneously, or meta-core reaches Tri-Critical	All organism alerts + all channel edges highlighted + meta-core inner triangle severely deformed with global instability pulsing + outer triangle border pulses amber/red.

9.2 Alert Propagation Logic

```
for each organism O in {Meridian, Verdara Ultra, HydroCore}:
  for each node N in O that crosses critical threshold:
    for each cross-node coupling (N → N_other, O_other):
      if coupling_strength(N, N_other) > threshold_minimum:
        activate_channel_pulse(channel(O, O_other))
        activate_cross_node_link_highlight(N, N_other)
        update_meta_core_with_cascade_flag(O, O_other)
```

The cascade flag on the meta-core produces a distinct visual indicator: a directional arrow overlaid on the inner triangle pointing from the originating organism’s axis toward the receiving organism’s axis — encoding both that a cascade is occurring and which direction it is flowing.

9.3 Cascade Suppression

Domain Integrity Principle: Alert propagation activates visual signals; it does not generate governance recommendations in the receiving organism unless the receiving organism’s own nodes independently confirm degradation. Each organism

governs its own domain based on its own node states. Cross-organism visual signals inform operators that coupling effects may be propagating, but they do not override the receiving organism's governance logic.

10 Interaction Modes — User Interface Specification

10.1 Default View

The default view renders the 2D Tri-Organism Geometry at Standard organism detail level with: all three organisms visible at their vertex positions; all three interaction channels rendered as semi-transparent bands with active flow lines; the central meta-core visible and current; no cross-node coupling links visible (available on demand); all active intra-organism alerts visible.

10.2 Hover Interactions

Hovering an organism: That organism expands to Detail level (full node labeling, state values, threshold indicators). The other two organisms revert to Overview level. That organism's cross-node coupling links activate at reduced opacity.

Hovering an interaction channel edge: The channel expands to show active cross-node coupling links for that edge. The specific bridging nodes in both organisms are highlighted. A tooltip shows the physical relationship description for the top 3 most active coupling pairs.

Hovering the meta-core: A summary overlay shows the tri-organism health table: H_{Meridian} , $H_{\text{VerdaraUltra}}$, $H_{\text{HydroCore}}$; current system state classification; top 3 most critical nodes across all three organisms; and active cascade flags with direction.

10.3 Selection Interactions

Selecting an organism: Locks that organism at Detail level. Shows a sidebar with full primitive state vector and active alerts. Maintains Standard level for the other two.

Selecting an interaction channel: Opens a panel showing the full cross-node coupling table for that channel pair. Highlights all currently active coupling links at full opacity. Shows a time series of coupling strength over the past 24 hours.

Selecting the meta-core: Opens a system-level dashboard showing all three organisms' primitive states in tabular format, tri-organism state classification history, and active cascade flags with originating node, receiving organism, and coupling strength.

10.4 Filter Controls

- **Toggle interaction channels:** Show or hide the channel bands between organisms
- **Toggle cross-node links:** Show or hide the coupling link overlay for a selected channel
- **Toggle meta-core:** Show or hide the central meta-core
- **Temporal mode:** Switch to time-stack 3D view for historical analysis
- **3D mode toggle:** Switch between 2D flat view and 3D stratified view
- **Alert-only mode:** Highlight only nodes, channels, and links currently in advisory or below; suppress optimal-state visual noise

11 Implementation Architecture

11.1 Data Architecture

The Tri-Organism Geometry requires a unified data layer that aggregates state from all three organisms and maintains the coupling computation:

```
{
  "tri_organism_state": {
    "timestamp": "ISO-8601",
    "organisms": {
      "meridian": { "primitives": [...], "nodes": [...] },
      "verdara_ultra": { "primitives": [...], "nodes": [...] },
      "hydrocore": { "primitives": [...], "nodes": [...] }
    },
    "meta_core": {
      "H_meridian": 0.0,
      "H_verdara": 0.0,
      "H_hydrocore": 0.0,
      "system_state": "Equilibrium|Advisory|Caution|Critical",
      "cascade_flags": []
    },
    "channels": {
      "meridian_verdara": { "strength": 0.0, "dominant_direction":
"M→V|V→M|balanced" },
      "verdara_hydrocore": { "strength": 0.0, "dominant_direction":
"V→H|H→V|balanced" },
      "hydrocore_meridian": { "strength": 0.0, "dominant_direction":
"H→M|M→H|balanced" }
    },
    "active_coupling_links": []
  }
}
```

11.2 Rendering Architecture

- **2D canvas rendering:** SVG or HTML Canvas with polar coordinate primitives for organism ring rendering
- **3D rendering:** WebGL or Three.js for the 3D stratified and tetrahedral views
- **Real-time state updates:** Organism state vectors update at each sensing cycle; the geometry must re-render within 100 ms of a state update
- **Interaction event handling:** Hover, select, filter, and temporal scrub interactions must be responsive within 50 ms

11.3 Determinism Guarantee

The Tri-Organism Geometry inherits the determinism guarantee of all Lume organisms: Given identical input state vectors from all three organisms, the geometry renders identically. No randomness, no animation timing that affects visual meaning, no layout choices that vary between renders. Two instances of the

geometry receiving the same state data must be visually identical. This extends to the meta-core computation, coupling strength calculations, cascade propagation logic, and alert classification. All are pure functions of the input state vectors.

12 Applications and Future Extensions

12.1 Immediate Application Contexts

- **Remote wilderness communities:** A community at the end of a mountain road with micro-hydro generation, outdoor infrastructure, and water distribution can use the Tri-Organism Geometry to monitor all three physical systems simultaneously and identify cross-domain cascades (drought → hydro generation loss → energy degradation → water treatment failure) before they fully develop.
- **Ultra-distance outdoor events:** A trail race or wilderness expedition using Verdara Ultra for route safety, HydroCore for water source management, and Meridian for communications infrastructure can use the Tri-Organism Geometry as the unified operational dashboard.
- **Watershed management with energy infrastructure:** A regional watershed management operation that includes hydroelectric generation, municipal water distribution, and outdoor safety governance can use the Tri-Organism Geometry as the unified operational awareness layer.

12.2 Extension to N Organisms

The triangular three-organism geometry is the minimal instantiation of a general N-organism polygon geometry. As the Lume organism ecosystem grows: four organisms use a square/rhombus arrangement; five organisms use a pentagon; N organisms use a regular N-gon with the meta-core at center. Each additional organism adds $(N-1)$ new pairwise interaction channels and a corresponding expansion of the meta-core axis structure. The coupling taxonomy (Appendix B) must be extended for each new organism pair.

12.3 BioCore Integration

BioCore — the internal biological flow organism — was excluded from the Tri-Organism Geometry because its domain is the internal human body rather than a co-located external physical domain. However, the human practitioner who operates within the outdoor and

water environments governed by Verdara Ultra and HydroCore, and who depends on the energy infrastructure governed by Meridian, is themselves a biological organism with a BioCore state.

A four-organism geometry that adds BioCore as a fourth vertex — or situates BioCore within the triangle (the practitioner embedded in the physical world) — is a natural extension that would enable a complete governance view of the practitioner-in-environment system. This extension is a subject for future work.

Appendix A — Organism Coordinate System Specification

Triangle side length: L			
Centroid: origin (0, 0)			
Vertex positions:			
Meridian:	(0,	+L × $\sqrt{3}/3$)	= (0, +0.5774L)
Verdara Ultra:	(-L/2,	-L × $\sqrt{3}/6$)	= (-0.5L, -0.2887L)
HydroCore:	(+L/2,	-L × $\sqrt{3}/6$)	= (+0.5L, -0.2887L)
Organism ring radii (default): 0.28L			
Interaction channel width (default): 0.12L			
Meta-core position: (0, 0) – centroid			
Meta-core axis base length (default): 0.08L			
3D Z-offsets (units of L):			
Meridian:	+0.5L		
Verdara Ultra:	0.0L		
HydroCore:	-0.5L		

Appendix B — Complete Cross-Node Coupling Table

The following tables list all formally specified cross-node couplings with their correlation coefficients. Coefficients represent the fraction of variance in the target node’s state explained by the source node’s state under normal operating conditions. All coefficients are initial estimates subject to empirical calibration.

Meridian → Verdara Ultra:

Source	Target	Relationship	Coeff.
Mesh health	W10 Safe Weather Window	Real-time weather data quality	0.65
Outage risk	B11 Rescue Accessibility	Comms coverage for rescue	0.80
Transmission stability	E4 Visibility (data)	Sensor network data quality	0.55

Verdara Ultra → Meridian:

Source	Target	Relationship	Coeff.
W4 Precipitation Type	Infrastructure health	Storm damage to exposed infrastructure	0.60
W2 Wind Speed	Transmission stability	Wind damage to lines and structures	0.65
E7 AQI (fire context)	Infrastructure health	Wildfire infrastructure threat	0.55

Verdara Ultra → HydroCore:

Source	Target	Relationship	Coeff.
W4 Precipitation Type	V7 Flood Probability	Storm type determines runoff	0.85
W5 Intensity	V1 Flow Rate	Rainfall drives flow	0.80
W5 Intensity	U1 Turbidity	Rain increases sediment	0.75
T4 Surface Stability	V5 Soil Absorption	Saturated soils reduce absorption	0.70
W1 Temperature	T3 Freeze Risk	Air temp drives water freeze	0.90
W1 Temperature	V8 Drought Probability	High temp accelerates evaporation	0.65
B7 Water Availability	V8 Drought Probability	Drought reduces outdoor water access	0.75
B4 Pathogen Risk	U4 Pathogen Index	Surface contamination correlates	0.70
T9 Water Crossing	V7 Flood Probability	Crossing difficulty reflects flood	0.85

HydroCore → Verdara Ultra:

Source	Target	Relationship	Coeff.
V7 Flood Probability	T9 Water Crossing	Flood makes crossings hazardous	0.85
V7 Flood Probability	T4 Surface Stability	Flood destabilizes terrain	0.70
V8 Drought Probability	B7 Water Availability	Drought reduces outdoor water sources	0.80
U4 Pathogen Index	B4 Pathogen Risk	Contaminated water is outdoor hazard	0.70
T3 Freeze Risk	T8 Traction Quality	Ice formation degrades traction	0.75

HydroCore → Meridian:

Source	Target	Relationship	Coeff.
V1 Flow Rate	Generation capacity	Hydro generation is flow-dependent	0.90
P4 Head Height	Generation capacity	Head determines power potential	0.85
P5 Hydraulic Gradient	Generation capacity	Gradient determines flow velocity	0.80
V8 Drought Probability	Generation capacity	Drought threatens hydro generation	0.85
V7 Flood Probability	Infrastructure health	Flood damages energy infrastructure	0.70

Meridian → HydroCore:

Source	Target	Relationship	Coeff.
Generation capacity	V1 Flow Rate (pump)	Power loss reduces pumping capacity	0.75
Distribution stability	U11 Potability Index	Power loss disrupts water treatment	0.70
Infrastructure health	P9 Pressure Stability	Power-dependent pressure control	0.60

Appendix C — Meta-Core Computation Specification

```

function compute_meta_core(meridian_state, verdara_state,
hydrocore_state):
    // Compute organism health scalars
    H_M = mean(meridian_state.primitives)
    H_V = mean(verdara_state.primitives)
    H_H = mean(hydrocore_state.primitives)

    // Compute axis lengths
    axis_M = base_length * (H_M + 1.0) / 2.0
    axis_V = base_length * (H_V + 1.0) / 2.0
    axis_H = base_length * (H_H + 1.0) / 2.0

    // Classify system state
    H_min = min(H_M, H_V, H_H)
    any_critical = any(node.state ≤ -0.8 for organism in all for node
in organism.nodes)

    if H_min ≥ 0.3 and not any_critical:
        system_state = "Tri-Equilibrium"
    elif H_min ≥ 0.0:
        system_state = "Tri-Advisory"
    elif H_min ≥ -0.5:
        system_state = "Tri-Caution"
    else:
        system_state = "Tri-Critical"

    // Compute inner triangle blend color
    w_M = max(0, H_M + 1.0) / 2.0
    w_V = max(0, H_V + 1.0) / 2.0
    w_H = max(0, H_H + 1.0) / 2.0
    total_w = w_M + w_V + w_H

    blend_color = (w_M/total_w × meridian_color
                  + w_V/total_w × verdara_color
                  + w_H/total_w × hydrocore_color)

    return MetaCore(axis_M, axis_V, axis_H, system_state,
blend_color)

```

Appendix D — Alert Propagation Logic

```

function propagate_alerts(organisms, coupling_table):
    cascade_flags = []

    for organism_src in organisms:
        for node_src in organism_src.nodes:
            if node_src.state ≤ -0.8: // Critical threshold
                for (node_tgt, organism_tgt, coefficient) in
coupling_table[node_src]:
                    strength = min(abs(node_src.state),
abs(node_tgt.state)) × coefficient

                    if strength > COUPLING_MINIMUM_THRESHOLD:
                        pulse_channel(channel(organism_src,
organism_tgt))

                        highlight_cross_node_link(node_src, node_tgt,
strength)

                        cascade_flags.append(CascadeFlag(
                            source_organism = organism_src,
                            source_node = node_src,
                            target_organism = organism_tgt,
                            target_node = node_tgt,
                            strength = strength
                        ))

    meta_core.cascade_flags = cascade_flags

    // System alert if cascade involves multiple organisms
    affected = set(f.source_organism for f in cascade_flags)
                | set(f.target_organism for f in cascade_flags)

    if len(affected) ≥ 2:
        activate_system_alert(affected, cascade_flags)

    return cascade_flags

```

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Patent Pending — Patent Pending.

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Lume-X: Provisionally patented deterministic control runtime.

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